

Arijit Dey | Curriculum Vitae

Second Year Graduate Student in Statistics – Duke University

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Education

Duke University Ph.D. in Statistics	Durham, USA 2024–Present
Indian Institute of Technology Kanpur (IITK) M.Sc. in Statistics, <i>CPI: 9.55/10</i>	Kanpur, India 2022–2024
Ramakrishna Mission Residential College (Autonomous), Narendrapur B.Sc. in Statistics, <i>CGPA: 9.58/10</i>	Kolkata, India 2019–2022

Publications

Peer-reviewed

- **A. Dey**, and A. Hazra, (2025). A Semiparametric Generalized Exponential Regression Model with a Principled Distance-based Prior. *Statistical Papers*. [Link](#)
- **A. Dey**, X. Yan, and B. Chakraborty (2026). Sample size calculation of SMART design with a skewed outcome: Application to the SMART+ study. *Statistical Methods in Medical Research*. [Link](#)

Submitted

- D. Munson, **A. Dey**, and S. Mak (2026+). Environmental policy in the context of complex systems: Statistical sensitivity testing and policy optimization using agent-based models. *Environmental Modelling and Software*.

Current Research Projects

Non-stationary Multivariate Spatial Skew-t Process for Extreme Precipitation Indices.....

Mentor: Dr. Arnab Hazra, Mathematics and Statistics department, IITK **Mar'26–Present**

- Developing a Bayesian multivariate spatial skew-t model to jointly analyze extreme precipitation indices across India using gridded rainfall data, by introducing a non-stationary correlation structure across spatial regions.

Q-Learning for SMART Designs with Censored Survival Outcomes.....

Mentor: Dr. Bibhas Chakraborty and Dr. Xiaoxi Yan, Duke-NUS Medical School **Feb'26–Present**

- Developing a semiparametric Q-learning framework for multi-stage clinical decision making under right-censored survival data, including custom R implementations with inverse-probability censoring weights and large-scale simulation studies to evaluate optimal treatment regimes.

Sensitivity and Regularity of Optimizer Maps in Bayesian Optimization.....

Mentor: Dr. Simon Mak, Dept. of Statistical Sciences, Duke University **Jan'26–Present**

- Developing theoretical guarantees for stability of optimal decisions in parametric black-box optimization problems of the form $f(x, \theta)$. The work characterizes continuity and smooth dependence of the optimizer on noise parameters and motivates curvature-aware robust acquisition strategies.

Experiences

○ Teaching Assistant, Duke University

- STA199 Introduction to Data Science and Statistical Thinking. Spring 25, Fall 25, Spring 26
- STA325 Machine Learning and Data Mining. Fall 26

○ Data Preparation Analyst, ASA DataFest — Duke University **Feb 26**

Prepared and validated datasets for the ASA DataFest data analysis competition.

- Research Assistant, IIT Kanpur
- Teaching Assistant, Online Degree Program, IIT Madras

Spring 24

Summer 23

Professional Service

- Ad hoc Reviewer for
 - *Technometrics*
 - *Journal of Applied Statistics: Environmental Statistics and Data Science*
 - Statistical Methods in Medical Research

Academic Achievements

- *N. Balakrishnan Award in Statistics* for the best final-year M.Sc. (2-year) statistics student, IIT Kanpur.
- *Academic Excellence Award* in Statistics for 2022 and 2023, IIT Kanpur.
- All India Rank 15 in Joint Admission Test for Masters (JAM) 2022 among 2912 candidates.
- *Swami Lokeshwarananda award* for best all-around performances in U.G. courses 2021-22.
- *DST-inspire scholarship* for being in the top 1 percentile of higher secondary board exams.

Relevant Coursework

- **PhD Level Course (Duke University)**
Bayesian Statistical Modelling, Probability and Measure Theory, Linear Models, Statistical Inference, Probability and Statistical Models, High-dimensional Statistics, Theory and Algorithms of Machine Learning
- **Masters Level Course (IIT Kanpur)**
Linear Algebra, Real Analysis, Complex Analysis, Probability and Measure Theory, Statistical Inference, Regression Analysis, Time Series Analysis, Multivariate Analysis, Stochastic Process, Statistical Computing, Non-parametric Inference, Game Theory, Data Science Lab I, II & III, Intro to Machine Learning.

Skills

Programming Expertise: *R, Python*

System software: $\text{\LaTeX}$, GitHub, Microsoft Office

Languages: English (Professional), Bengali (Native), Hindi (Limited), Spanish (Beginner)

Extra Curricular

Positions of Responsibility:

- Secretary of *Raktarpan*, a club under *IITK* with the responsibility of organizing blood donation camps.
- General Secretary of the *Student's Council*, RKMRC Narendrapur, 2021-22.
- Convenor of the *Sports Subcommittee*, RKMRC Narendrapur, 2021-22.
- Junior Under Officer (JUO), *NCC wing* of RKMRC Narendrapur, 2021-22.

Sports: Active participation in games including Volleyball, Tennis, Football, and Cricket.